

A daraja — Array va Slice

1-masala: Slice bilan o'rtacha qiymat

Butun sonlardan iborat slice bering: `[]int{12, 7, 25, 3, 18}`.

- Slice ichidagi barcha sonlar yig'indisini hisoblang.
- O'rtacha (`average`) qiymatni toping.
- `len()` dan foydalaning, sonlar sonini qattiq (hardcoded) yozmang.

2-masala: Slice'ni o'stirish

Bo'sh slice (`var s []int`) yarating va `append` yordamida 1 dan 10 gacha sonlarni qo'shing. So'ng faqat **juft** sonlarni yangi slice'ga ajratib oling va chiqaring.

Tekshiriladi: `append`, `for` sikli, `%` operatori, slice'ni filtrlash.

B daraja — Map

3-masala: So'zlarni sanash

Quyidagi slice berilgan:

```
mevalar := []string{"olma", "anor", "olma", "uzum", "anor", "olma"}
```

`map[string]int` yarating va har bir meva nechta marta uchraganini sanang.

Natija: `olma: 3, anor: 2, uzum: 1`.

4-masala: Telefon kitobchasi

`map[string]string` ko'rinishida telefon kitobchasi yarating (ism → raqam).

- Kamida 3 ta kontakt qo'shing.
- Berilgan ism bo'yicha raqamni qidiring va **topilmasa** "Topilmadi" deb chiqaring

(`value, ok := m[key]` shaklini ishlatib).

- Bitta kontaktni `delete` bilan o'chiring.

Tekshiriladi: map yaratish, `comma-ok` idiomasi, `delete`.

C daraja — Struct

5-masala: Talaba structi

`Student` nomli struct yarating: `Ism string`, `Yosh int`, `Baho float64`.

- 3 ta talaba yarating va ularni `[]Student` slice'ga joylang.
- Slice bo'ylab aylanib (`for range`), bahosi 4.0 dan yuqori talabalarni chiqaring.

6-masala: Method bilan struct

5-masaladagi `Student` ga method qo'shing:

```
func (s Student) Otdimi() bool { ... }
```

Agar baho 3.0 dan baland bo'lsa `true` qaytarsin. Har bir talaba uchun

`talaba.Otdimi()` ni chaqirib natijani chiqaring.

Tekshiriladi: method receiver, value vs pointer tushunchasi.

7-masala: Ichma-ich (nested) struct

`Manzil` structi yarating (`Shahar`, `Kocha`). Keyin `Student` ichiga

`Manzil` ni maydon sifatida joylang. Bitta talaba yaratib, uning shahrini

`talaba.Manzil.Shahar` orqali chiqaring.

D daraja — Interface (eng muhim qism)

8-masala: Shakl yuzasi (klassik)

`Shape` (Shakl) interfeysini yarating:

```
type Shape interface {
    Yuza() float64
}
```

- `Doira` (radius bilan) va `Tortburchak` (eni va bo'yi bilan) structlarini yarating.
- Ikkalasi ham `Yuza()` methodini amalga oshirsin.
- `[]Shape` slice yaratib, har bir shaklning yuzasini bitta `for` sikli ichida chiqaring.

Bu masala interfeysni **chinakam tushunganini** ko'rsatadi — har xil structlarni

bitta interface orqali bir xil ishlata olish.

9-masala: `interface{}` va tur tekshiruvi

`[]interface{}{42, "salom", 3.14, true}` slice'ini bering.

Har bir element bo'ylab aylaning va `switch v := x.(type)` (type switch) yordamida har bir elementning **turini aniqlab**, mos xabar chiqaring.

Masalan: `42 — bu butun son (int)`.

Tekshiriladi: empty interface, type assertion / type switch.

E daraja — Hammasini birlashtirish (yakuniy)

10-masala: Mini do'kon tizimi

Quyidagilarni birlashtiring:

- `Mahsulot` structi: `Nomi string`, `Narxi float64`, `Soni int`.
- `[]Mahsulot` slice — kamida 4 ta mahsulot.
- `map[string]float64` — mahsulot nomi → umumiy qiymati (`Narxi * Soni`).
- Eng qimmat mahsulotni topadigan funksiya yozing.
- `Qimmatbaho() bool` methodi: narxi 100000 dan oshsa `true`.

Hammasini `main` ichida ishlatib, do'kon hisobotini chiroyli chiqaring.

Bu masalani toza yecha olsa — o'quvchi struct, slice, map va method'larni

amalda erkin qo'llay oladi degani.